AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 25 · STUDENT LESSON

# Running the primary experiment

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

## Follow the frozen experiment and retain every eligible outcome

Report completion coverage, paired failures and later revisions without hiding missing cases.

**Keep one habit:** Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

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MAIN LESSON · SECTION 1

## Follow the frozen experiment and retain every eligible outcome

> Report completion coverage, paired failures and later revisions without hiding missing cases.

### Learning goals

I can explain the **mechanism** and terminology of running the primary experiment.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

## Starting explanation

- Recall Reproducible environment and records. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

## End goal

Explain running the primary experiment using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

**Your learning target**

**Remember before reading**

This week you will practise:

Explain the **mechanism** and terminology of running the primary experiment

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

**Retrieve what you already know**

**This week’s end goal**

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MAIN LESSON · SECTION 2

# Understand the question and evidence contract

> Report completion coverage, paired failures and later revisions without hiding missing cases.

## Start with the research problem

The primary run follows the recorded plan rather than changing rules to favor the result. Capture correct, incorrect and unassessed outcomes separately. When the plan reports a completed-case rate, show the completed denominator and coverage of all eligible cases. Unassessed cases are not silently successes or failures.

If inspecting a failure leads to a revised model, preserve the frozen result and label same-set revision results as development evidence.

## Original concept explanation

The primary experiment follows the frozen plan and captures every eligible outcome. Unexpected failures are recorded rather than silently retried away.

### Complete case inputs, rules and limits

| Primary-run item | Complete supplied exercise contract |
|---|---|
| Main eligible cases | Twenty |
| Observed categories | Sixteen pass, two fail, two unassessed after outage |
| Planned performance metric | Passes divided by completed cases |
| Planned coverage metric | Completed divided by eligible cases |
| Retries and stopping | Follow the stated rule; no extra retry policy supplied here |
| Separate picture | Frozen A and B on the same six cases, binary correctness supplied |

| Primary-run item | Complete supplied exercise contract |
| --- | --- |
| After inspecting case four | Tuning B creates a later revision; new outputs are not supplied |
| Scope | Supplied fictional records, not a new model run or population-advantage claim |

### Name given and missing evidence

**List the given facts and one fact that is not supplied.**

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

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MAIN LESSON · SECTION 3

## Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

### Original declared procedure

Run declared cases, store raw paired results and deviations, and keep primary analysis separate from exploratory repairs. Stop or escalate only according to the stated rule.

### Complete main-case evidence ledger

| Main quantity | Calculation | Interpretation |
|---|---|---|
| Eligible | 16 + 2 + 2 = 20 | All planned eligible cases |
| Completed | 16 + 2 = 18 | Assessed pass/fail outcomes |
| Completed-case pass rate | 16/18 = 8/9 ≈ 88.889% | Rate among completed cases |
| Coverage | 18/20 = 90% | Two of twenty remain unassessed |
| Unassessed | 2 | Retain as unknown outcomes |
| Different possible metric | 16/20 = 80% known passes among all eligible | Not the planned completed-case rate |
| Inference limit | Missingness may matter | Do not assume unassessed outcomes resemble completed cases |

## State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

**Running the primary experiment — evidence trace**

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Twenty eligible cases: 16 pass, 2 fail, 2 unassessed after outage. Plan reports completed rate plus coverage. | Run declared cases, store raw paired results and deviations, and keep primary analysis separate from exploratory repairs. Stop or escalate only according to the stated rule. | 16/18≈88.889% completed pass; coverage 18/20=90%; two unassessed remain. No 20/20 assessment claim. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

**Connect the picture and the explanation**

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MAIN LESSON · SECTION 4

# Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

## Main task and response

- Twenty eligible cases: 16 pass, 2 fail, 2 unassessed after outage. Plan reports completed rate plus coverage. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

### Predict the pictured change

- Change: Tune B after inspecting its case-four failure, then rescore the same set.

▼ Check after predicting

**Prediction**: Label the rerun as development on exposed cases, not untouched primary evaluation.

Reason: The evaluation outcomes have now influenced the version.

▼ Earlier wording, alternatives and picture transcripts

**Compare a familiar case with a changed case**

**Change one thing in the pictured case**

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MAIN LESSON · SECTION 6

## Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

### Main case and original checked reasoning

Twenty eligible cases: 16 pass, 2 fail, 2 unassessed after outage. Plan reports completed rate plus coverage.

16/18≈88.889% completed pass; coverage 18/20=90%; two unassessed remain. No 20/20 assessment claim.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

## Separate pictured inputs and complete explanation

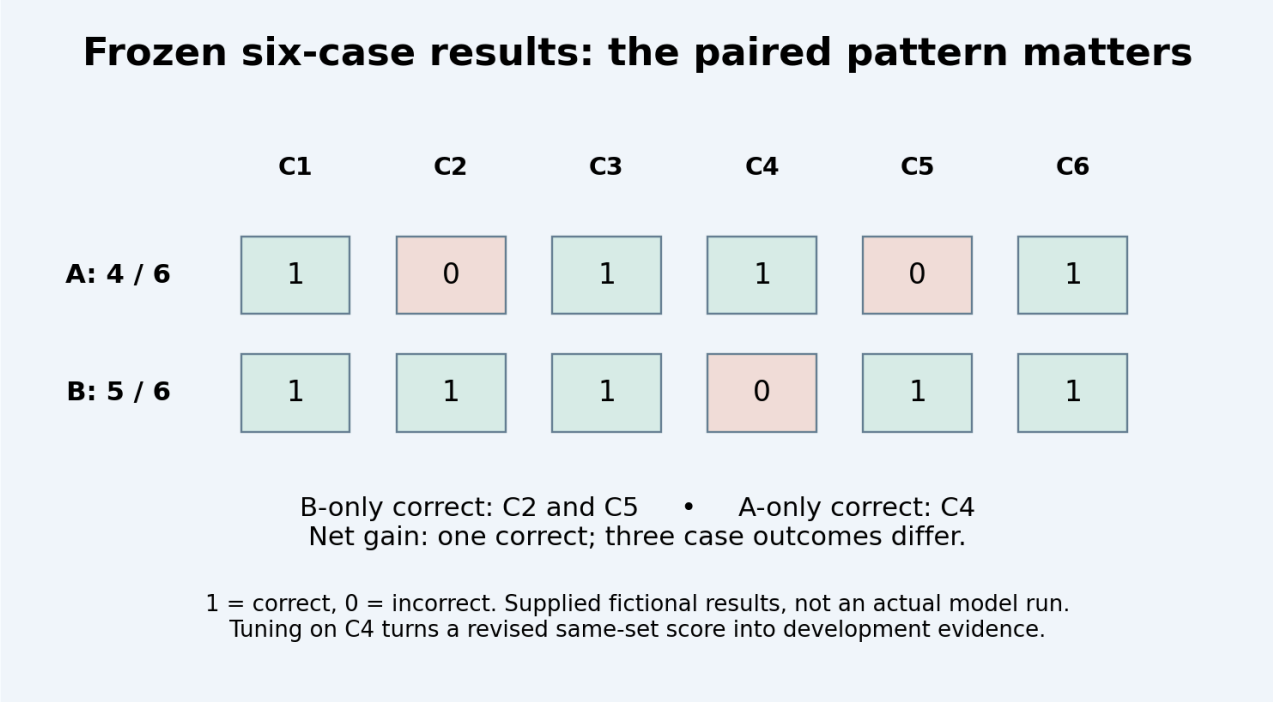
Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**.

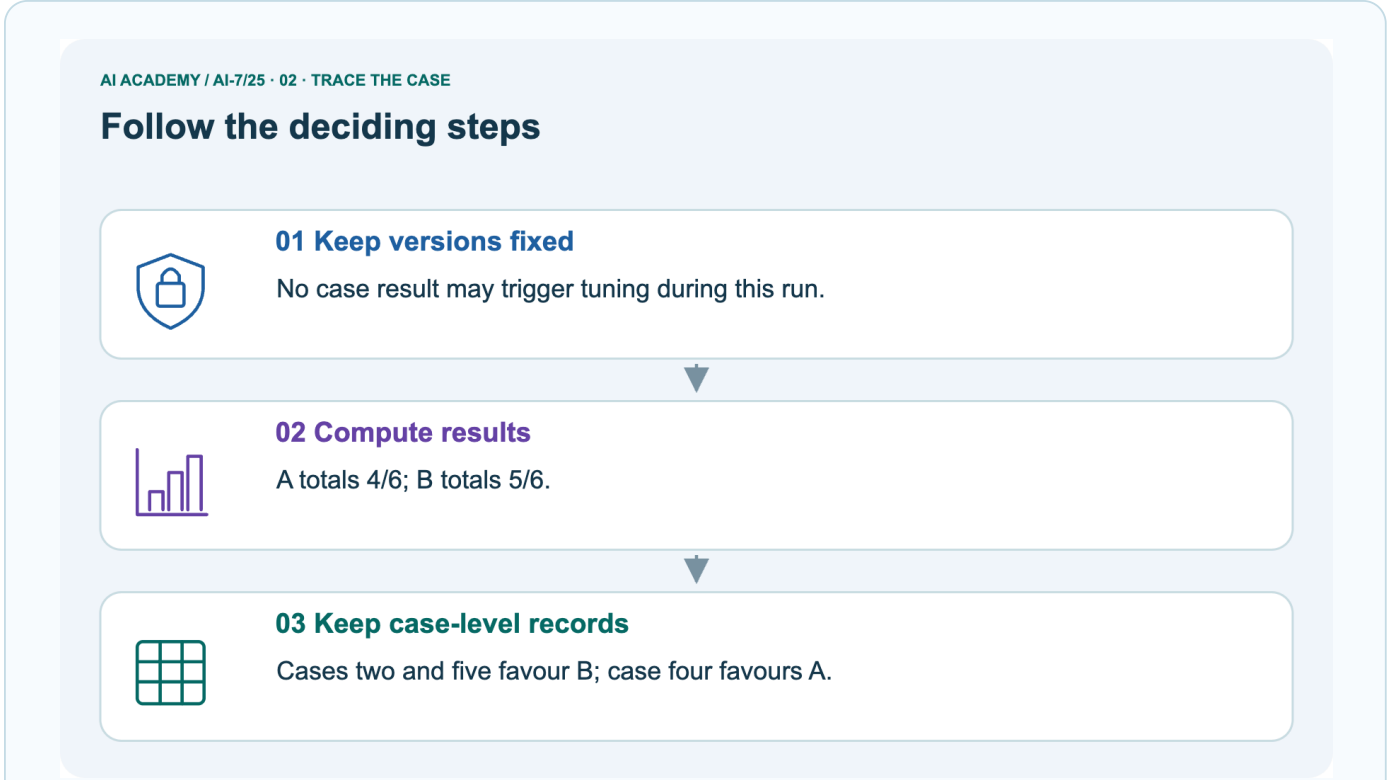
A scores [1,0,1,1,0,1]; B scores [1,1,1,0,1,1]. Both answer the same six cases. Binary entries mean 1=correct and 0=incorrect.

The separate six-case picture supplies A = [1, 0, 1, 1, 0, 1] and B = [1, 1, 1, 0, 1, 1], where one means correct. A has four correct and B five. Cases C2 and C5 favor B; C4 favors A; C1, C3 and C6 are ties. Thus the net gain of one correct hides three changed case outcomes. Tuning B after inspecting C4 creates a post-inspection version.

Rescoring that version on the same set is development evidence, not untouched primary confirmation. No revised outputs, statistical significance or population advantage are supplied.

## Read the labelled representation






Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Keep versions fixed | No case result may trigger tuning during this run. |
| 2. Compute results | A totals 4/6; B totals 5/6. |
| 3. Keep case-level records | Cases two and five favour B; case four favours A. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Running the primary experiment means following the declared protocol and retaining raw outcomes. Overall scores alone hide different failure patterns.

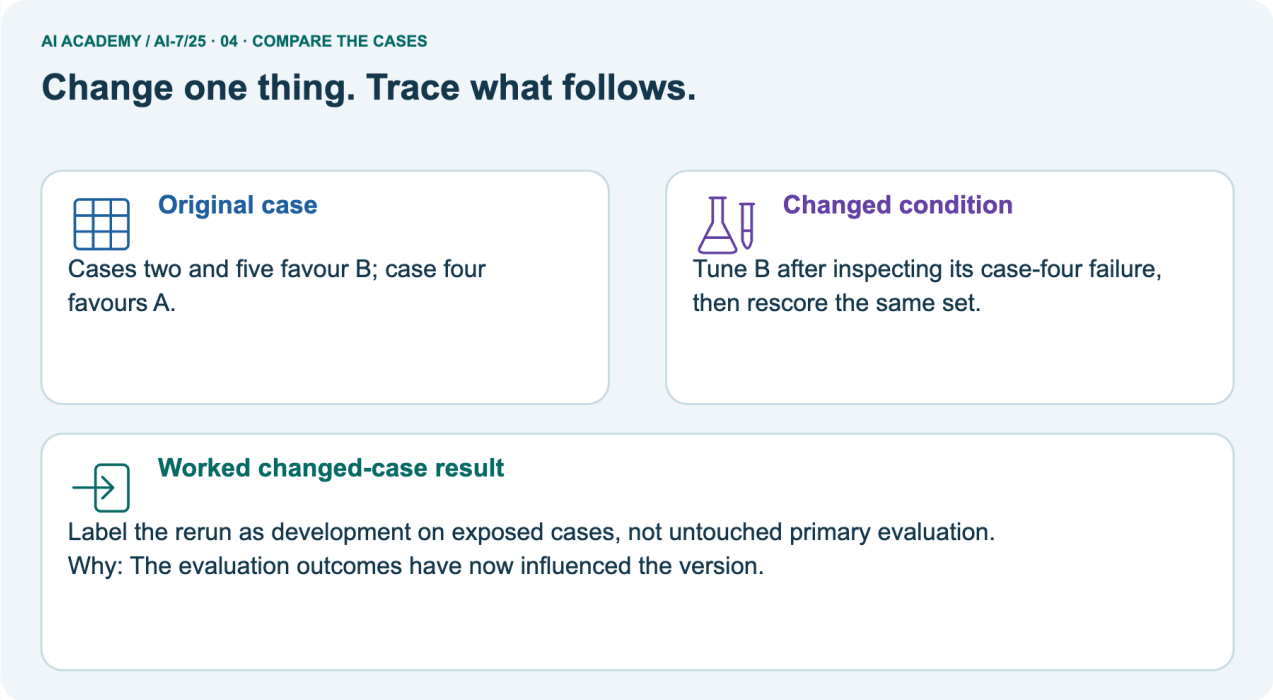
### Complete picture evidence and limits

| Paired case | A correct flag | B correct flag | B – A |
|---|---|---|---|
| C1 | 1 | 1 | 0 |
| C2 | 0 | 1 | 1 |
| C3 | 1 | 1 | 0 |
| C4 | 1 | 0 | -1 |
| C5 | 0 | 1 | 1 |
| C6 | 1 | 1 | 0 |
| Total | 4/6 | 5/6 | Net +1; three cases differ |
| B tuned after C4 | Frozen original retained | Revised score unknown | Same-set revision is development evidence |

## Predict the changed condition

- Change one thing: Tune B after inspecting its case-four failure, then rescore the same set. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting




Follow the labelled picture with the matching explanation. Open the vector diagram

## Explain what remains unknown

- Six illustrative cases do not establish a population advantage or statistical significance.
- Read the labelled picture: Compute A's and B's six-case counts and identify the cases that favour each. How must a B revision tuned on case four be labelled when evaluated again on the same set?

▼ Earlier wording, alternatives and picture transcripts

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Running the primary experiment

### The facts we will use

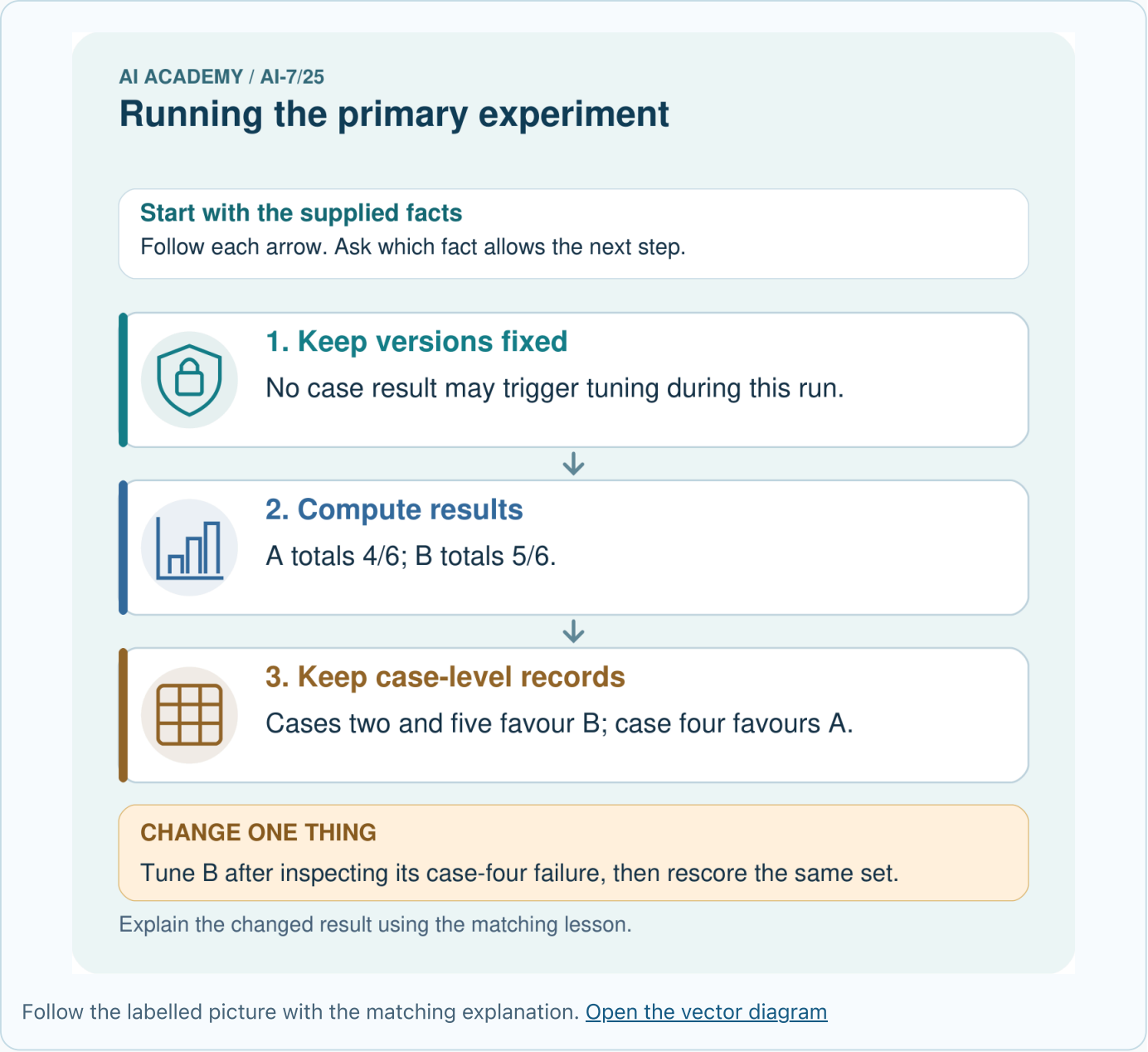
AI ACADEMY / AI-7/25

**Paired primary case outcomes**

| | Case 1 | Case 2 | Case 3 | Case 4 | Case 5 | Case 6 |
|---|---|---|---|---|---|---|
| Version A | 1 | 0 | 1 | 1 | 0 | 1 |
| Version B | 1 | 1 | 1 | 0 | 1 | 1 |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram




Follow the labelled picture with the matching explanation. Open the vector diagram

**Why this result follows**

**What this example does not establish**

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MAIN LESSON · SECTION 7

## Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

## Original qualification and repair setup

Following the plan and reporting deviations are required even when results contradict the hypothesis.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

## Complete boundary and misconception probes

| Primary-report audit | Checked interpretation |
| --- | --- |
| Two unassessed cases | Keep count explicit and outcomes unknown |
| Completed rate 16/18 | Not a claim all twenty were assessed |
| Known-pass/all-eligible rate 16/20 | A different denominator and metric |
| Net picture gain of one | Three paired outcomes differ |
| B correct count is higher | No population or significance claim follows from six cases alone |
| Inspect failure then tune B | Preserve frozen run and label revision separately |
| Same-set rescoring | Development evidence after tuning, not untouched confirmation |
| Retry until successful | Cannot silently replace original failures; retain attempts under the plan |

## Write and justify the repair

- A report omits this qualification: Following the plan and reporting deviations are required even when results contradict the hypothesis.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

**Locate the earliest unsupported step**

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MAIN LESSON · SECTION 8

## Practise the mechanism with progressively less help

> Keep the selected independent case for a first attempt.

### Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**Explain your prediction to a partner before comparing results.**

## Supported workbook trace

Use workbook W2 for the connected example “Running the primary experiment”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Individual scope check

- Use the worked case for Running the primary experiment. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

## Picture response

- Compute A’s and B’s six-case counts and identify the cases that favour each. How must a B revision tuned on case four be labelled when evaluated again on the same set?

**My explanation using a pictured step and the given facts**

### Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

**Practise with less help: Trace the worked case**

**Check your reasoning**

**Explain the picture**

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MAIN LESSON · SECTION 9

## Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

### Current investigation

Twenty eligible cases: 16 pass, 2 fail, 2 unassessed after outage. Plan reports completed rate plus coverage. Show the intermediate steps and give the conclusion with its scope.

### Complete investigation instructions

1. Predict the completed-case pass rate and coverage from the supplied 16/2/2 counts. The complete code checks nonnegative count totals and keeps zero-completion rates unknown rather than dividing by zero. Then trace all six paired picture rows, identifying C2/C5 for B and C4 for A. No model is run and no outage is induced. Preserve the original B result and leave revised results unknown because they were not supplied.

   Keep the twelve-case independent completion task for a first attempt.

```
"""Analyze stipulated primary-study records; no actual model experiment."""
from fractions import Fraction

def completion_summary(passes,failures,unassessed):
 values=(passes,failures,unassessed)
 if any(type(x) is not int or x<0 for x in values) or sum(values)==0:raise ValueError('nonnegative
integer counts and positive eligible total required')
```

```
 completed=passes+failures;eligible=completed+unassessed
 return
{'passes':passes,'failures':failures,'unassessed':unassessed,'completed':completed,'eligible':eligible,
         'completed_pass_rate':Fraction(passes,completed) if completed else None,
         'coverage':Fraction(completed,eligible)}
main=completion_summary(16,2,2)
assert main['completed_pass_rate']==Fraction(8,9) and main['coverage']==Fraction(9,10)
A=[1,0,1,1,0,1];B=[1,1,1,0,1,1]
paired=[{'case_id':f'C{i+1}','A':a,'B':b,'B_minus_A':b-a} for i,(a,b) in enumerate(zip(A,B))]
a_only=[r['case_id'] for r in paired if r['A']==1 and r['B']==0]
b_only=[r['case_id'] for r in paired if r['A']==0 and r['B']==1]
assert sum(A)==4 and sum(B)==5 and a_only==['C4'] and b_only==['C2','C5']
assert completion_summary(0,0,2)['completed_pass_rate'] is None
print('Main supplied completion summary:',main)
print('Separate frozen six-case picture:',paired)
print('A-only correct:',a_only,'B-only correct:',b_only,'net B-A:',sum(B)-sum(A))
print('Net one extra correct is not one changed case: three paired cases differ. Unassessed cases
remain unknown.')
print('A version tuned after inspecting C4 is a post-inspection revision. Same-set rescoring is
development evidence; its new outputs are not supplied.')
print('No model is run, no outage is induced, no significance or population advantage is
established, and a revised score is not invented.')
```

## Evidence record and reflection

| Record | Your evidence |
|---|---|
| Given input and fixed rule | |
| Prediction before checking | |
| Actual result, or “not run” | |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

**Predict, investigate and preserve the evidence**

**Plan → try → check**

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MAIN LESSON · SECTION 14

# Try the selected new case independently

> Show completed-case performance, assessment coverage and the unassessed count with their different denominators.

### Complete independent task

Twelve eligible cases include 9 passes, 1 failure and 2 unassessed cases. Report completed-case pass rate (passes / completed cases), assessment coverage (completed / eligible cases) and the unassessed count.

## First attempt and support record

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

### Independent case: Apply a new case independently

### Attempt this without hints

Twelve eligible cases include 9 passes, 1 failure and 2 unassessed cases. Report completed-case pass rate (passes / completed cases), assessment coverage (completed / eligible cases) and the unassessed count.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

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MAIN LESSON · SECTION 11

## Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

### Project task and artifact

Add a labelled running the primary experiment **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of running the primary experiment.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
|---|---|
| Component and version | |

| Project evidence | Record |
| --- | --- |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

## Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

**Evidence for your project**

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MAIN LESSON · SECTION 16

# Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

## Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of running the primary experiment

**My deciding step, evidence and one remaining question**

## Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

## Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

### Vocabulary and your explanation

| Word | Meaning |
|---|---|
| Primary run | The experiment under the declared main plan |
| Unassessed case | An eligible case without the required outcome evidence |
| Deviation log | A record of changes or interruptions |

**Use two terms to explain a step in this week's example.**

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

## Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

### Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Following the plan and reporting deviations are required even when results contradict the hypothesis.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

#### Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the research reasoning and its limits**

REFERENCE · SECTION 12

## Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

## Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

**My limited claim and a useful next check**

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

# Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

## Complete trace and evidence

### Complete evidence record

#### Running the primary experiment — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Twenty eligible cases: 16 pass, 2 fail, 2 unassessed after outage. Plan reports completed rate plus coverage. | Run declared cases, store raw paired results and deviations, and keep primary analysis separate from exploratory repairs. Stop or escalate only according to the stated rule. | 16/18≈88.889% completed pass; coverage 18/20=90%; two unassessed remain. No 20/20 assessment claim. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ **Reference · The earlier route checklist**

## Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

### Targets,route and first prediction

Start with this question: Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**.

| By the end, I can | How I will show it |
|---|---|
| Explain the mechanism and terminology of running the primary experiment | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

## Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

### First supported question and optional hint

- Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**.

- Which labelled step matches this detail? Cases two and five favour B; case four favours A.
- 1. Compute results
- 2. Keep versions fixed
- 3. Keep case-level records
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Keep case-level records Cases two and five favour B; case four favours A.

Running the primary experiment means following the declared protocol and retaining raw outcomes. Overall scores alone hide different failure patterns.

## Changed-case question and optional hint

- Changed-case practice: Tune B after inspecting its case-four failure, then rescore the same set. Which conclusion is supported by the supplied facts?
- 1. Label the rerun as development on exposed cases, not untouched primary evaluation.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Label the rerun as development on exposed cases, not untouched primary evaluation. The evaluation outcomes have now influenced the version.

## Repair and retain the record

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

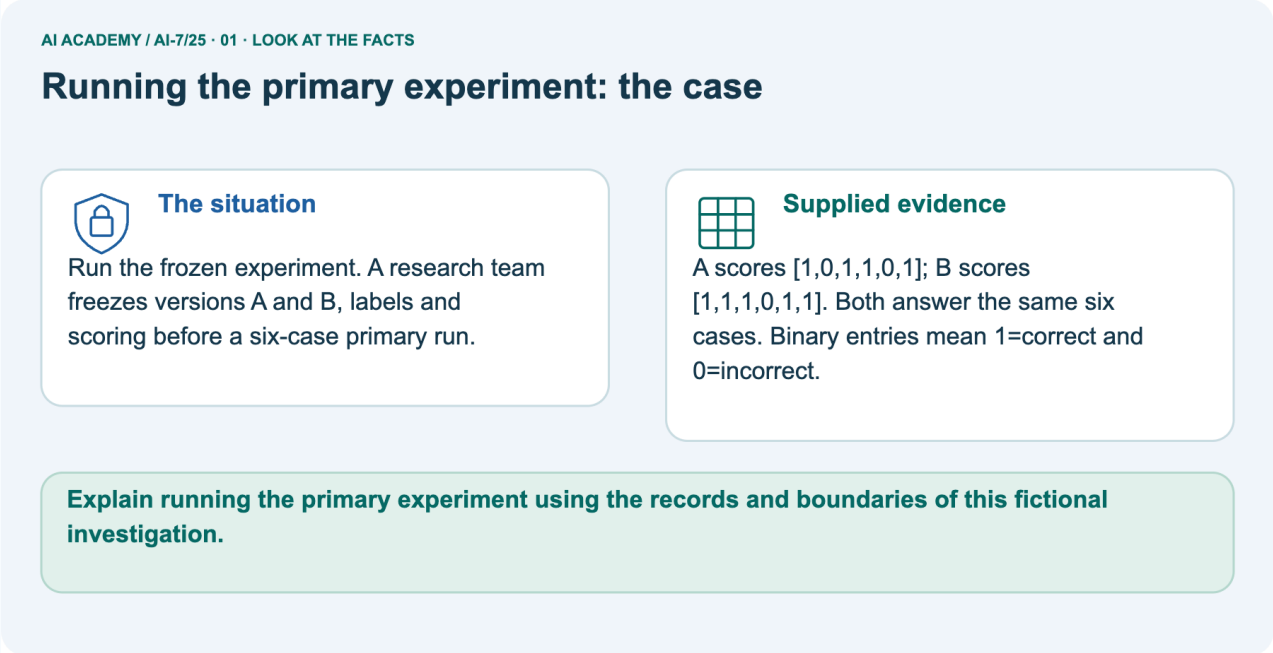
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

# Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

## 01. Running the primary experiment: the case



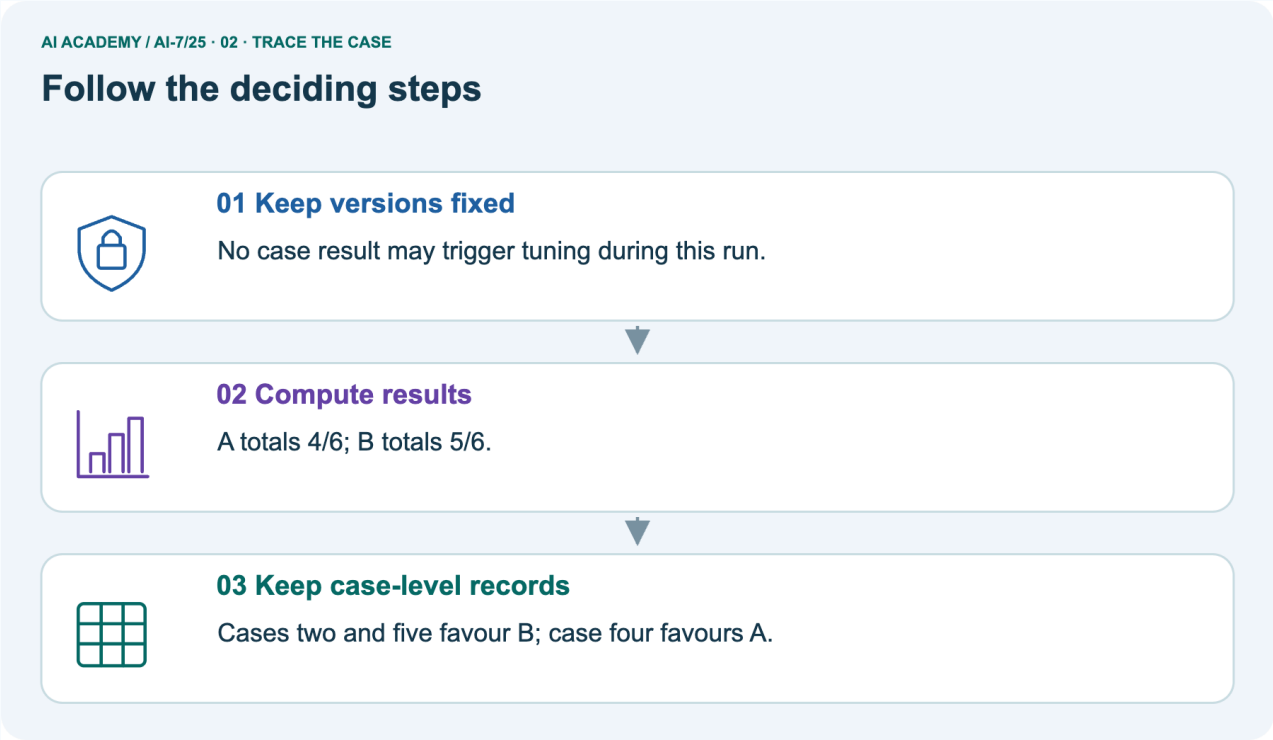

Follow the labelled picture with the matching explanation. Open the vector diagram

Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**. Supplied facts: A scores [1,0,1,1,0,1]; B scores [1,1,1,0,1,1]. Both answer the same six cases.

Binary entries mean 1=correct and 0=incorrect.

Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Keep versions fixed: No case result may trigger tuning during this run. → Compute results: A totals 4/6; B totals 5/6. → Keep case-level records: Cases two and five favour B; case four favours A.

Point to each numbered stage and explain which recorded **input** it uses.

## 03. Paired primary case outcomes

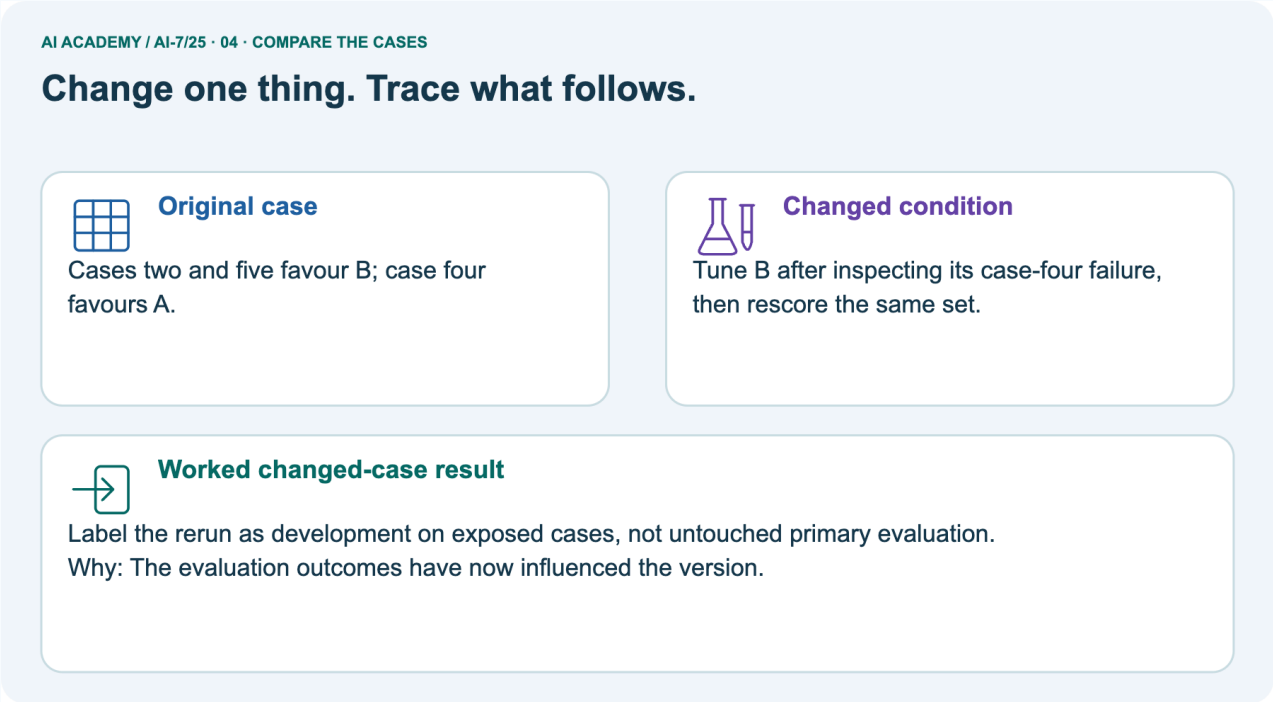



Follow the labelled picture with the matching explanation. Open the vector diagram

Paired primary case outcomes. Version A: Case 1 = 1, Case 2 = 0, Case 3 = 1, Case 4 = 1, Case 5 = 0, Case 6 = 1. Version B: Case 1 = 1, Case 2 = 1, Case 3 = 1, Case 4 = 0, Case 5 = 1, Case 6 = 1.

Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



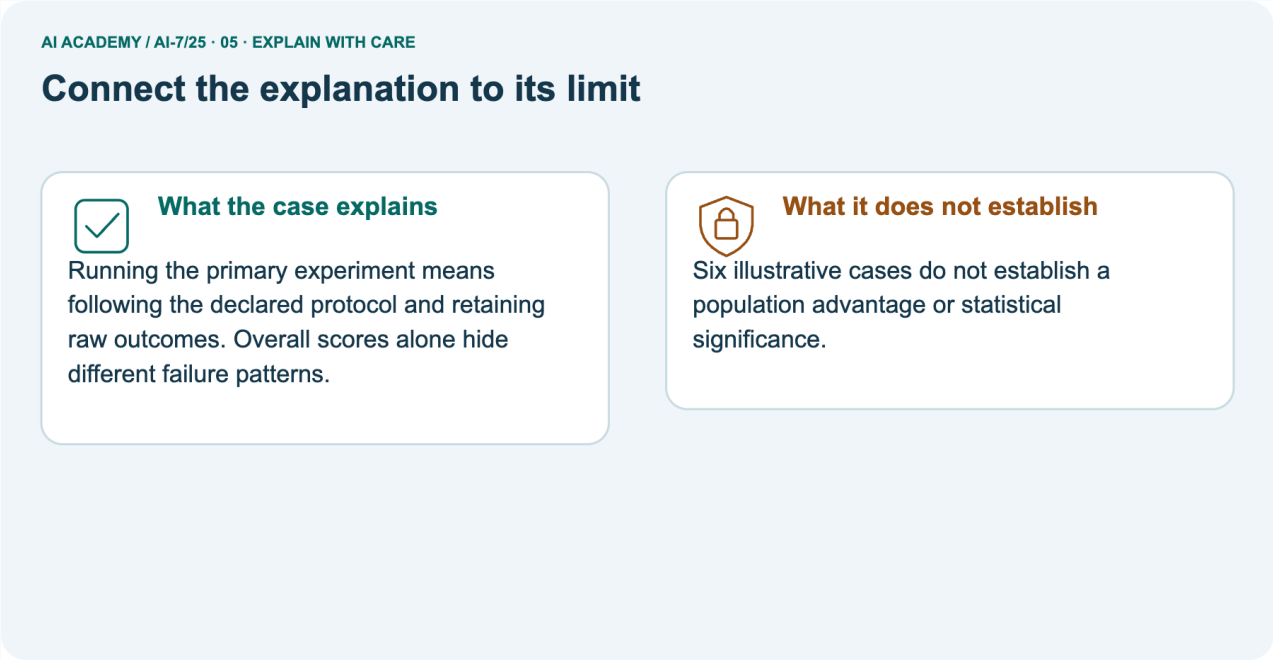

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Cases two and five favour B; case four favours A. Changed condition: Tune B after inspecting its case-four failure, then rescore the same set.

Expected result: Label the rerun as development on exposed cases, not untouched primary evaluation. Reason: The evaluation outcomes have now influenced the version.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit



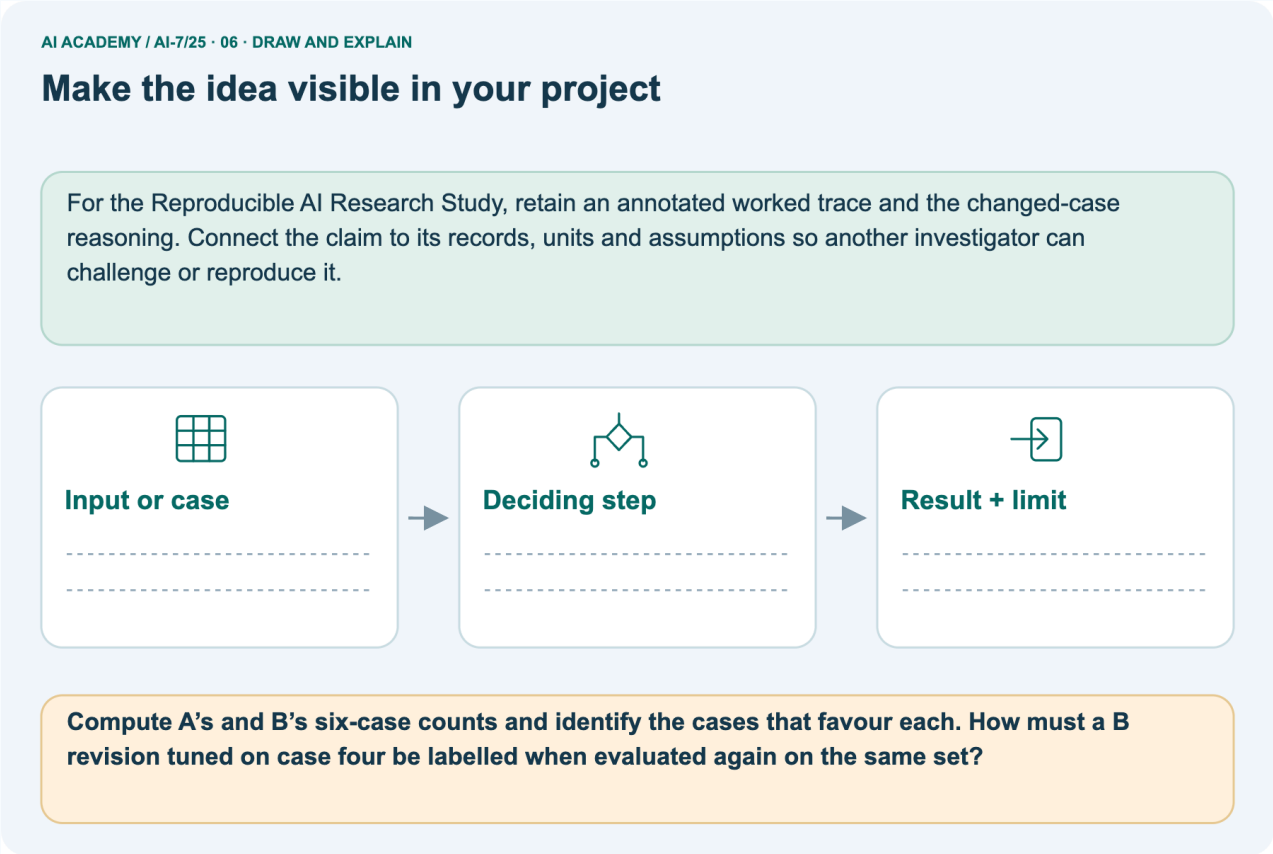

Follow the labelled picture with the matching explanation. Open the vector diagram

Running the primary experiment means following the declared protocol and retaining raw outcomes. Overall scores alone hide different failure patterns.

Limit: Six illustrative cases do not establish a population advantage or statistical significance.

Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?

Draw your project trace and justify the deciding step using **evidence**.

## ▼ Earlier wording, alternatives and picture transcripts

**01. Running the primary experiment: the case**

**02. Follow the deciding steps**

**03. Paired primary case outcomes**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.